

TOCUMEN INTL, Panama

WMO: 787920

Lat: 9.071N

Lon: 79.383W

Elev: 41

StdP: 100.83

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 20.9	(c) 21.8	(d) 16.9	(e) 12.1	(f) 29.9	(g) 17.8	(h) 12.9	(i) 28.9	(j) 7.7	(k) 29.3	(l) 6.7	(m) 29.8	(n) 0.2	(o) 310	(p) 0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 9.1	(c) 34.2	(d) 25.4	(e) 33.8	(f) 25.3	(g) 33.1	(h) 25.0	(i) 27.7	(j) 31.2	(k) 27.2	(l) 30.9	(m) 26.9	(n) 30.6	(o) 4.0	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	29.6	26.2	21.7	29.1	26.0	21.5	29.0	88.6	31.3	86.4	31.3	84.9	30.8	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.8	(b) 6.6	(c) 5.7	(e) 19.1	(f) 35.7	(g) 3.0	(h) 1.0	(i) 16.9	(j) 36.4	(k) 15.1	(l) 36.9	(m) 13.4	(n) 37.5	(o) 11.2	(p) 38.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			17.5	28.8	3.0	1.0	15.3	29.6	13.6	30.1	11.9	30.7	9.6	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	27.5	27.8	28.2	28.4	27.9	27.6	27.7	27.6	27.3	27.1	27.1	27.4
	DBStd	1.12	0.98	0.98	1.08	1.01	1.20	1.04	1.17	1.10	0.99	0.95	1.03	1.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6434	541	497	564	552	555	528	549	546	519	530	513	539
	CDD18.3	3393	283	264	306	302	297	278	291	288	269	272	263	281
	CDH23.3	33684	2909	2856	3464	3384	3031	2708	2838	2746	2454	2397	2235	2663
	CDH26.7	12333	1272	1273	1578	1501	1059	857	952	880	701	624	639	997
Wind	WSAvg	1.7	2.1	2.3	2.3	1.9	1.3	1.2	1.3	1.3	1.4	1.6	1.4	1.8
Precipitation	PrecAvg	2458	34	13	21	98	305	287	256	273	299	371	328	176
	PrecMax	3819	168	54	68	317	474	535	396	428	494	580	657	553
	PrecMin	1895	4	4	0	3	132	125	137	152	163	270	205	41
	PrecStd	402	37	12	21	68	75	97	60	59	85	84	110	128
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.1	34.9	35.1	34.2	34.0	34.1	34.1	33.2	33.0	33.1	33.6
		MCWB	25.1	24.4	24.8	25.4	26.4	27.0	26.4	26.6	26.5	26.8	26.5	26.0
	2%	DB	33.1	33.2	33.9	34.1	33.1	32.8	33.0	32.9	32.0	31.7	32.0	32.9
		MCWB	24.3	24.0	24.3	25.1	26.1	26.5	26.3	26.6	26.2	26.4	25.9	25.6
	5%	DB	32.2	32.8	33.1	33.2	32.1	31.7	32.0	31.8	31.0	30.8	31.1	32.0
		MCWB	24.1	23.9	24.1	24.8	26.0	26.3	26.2	26.4	26.2	26.2	25.9	25.1
	10%	DB	31.9	32.1	32.4	32.7	31.1	30.8	31.0	30.8	30.1	29.8	30.1	31.2
		MCWB	24.0	23.8	23.9	24.7	26.0	26.2	26.2	26.2	25.9	25.8	25.7	25.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.4	27.1	27.7	28.0	27.7	27.8	28.3	27.7	27.6	27.5	27.5
		MCDB	31.8	31.3	31.9	31.9	31.1	31.7	30.9	31.3	30.6	30.8	31.1	31.4
	2%	WB	25.9	25.6	25.9	26.7	27.2	27.1	27.1	27.3	26.9	26.8	26.7	26.7
		MCDB	30.9	30.7	31.1	31.0	30.8	31.0	30.8	30.7	30.1	29.9	30.2	30.6
	5%	WB	25.3	25.0	25.2	26.1	26.7	26.6	26.6	26.7	26.4	26.3	26.3	26.2
		MCDB	30.5	30.5	31.0	30.5	30.3	30.2	30.3	30.2	29.5	29.1	29.4	29.9
	10%	WB	24.7	24.5	24.7	25.7	26.2	26.2	26.2	26.2	26.1	26.0	26.0	25.8
		MCDB	30.0	30.2	30.6	30.1	29.7	29.4	29.7	29.5	29.0	28.6	28.9	29.4
Mean Daily Temperature Range	5% DB	MDBR	9.7	9.6	9.6	9.1	7.0	6.6	7.1	7.0	6.5	6.2	6.8	8.2
		MCDBR	10.2	10.4	10.4	10.2	8.6	8.1	8.4	8.4	7.8	7.6	8.2	9.3
		MCWBR	3.7	3.5	3.4	3.7	3.5	3.6	3.7	3.8	3.5	3.4	3.6	3.8
	5% WB	MCDBR	9.7	9.5	9.7	8.9	7.4	7.3	7.5	7.6	7.0	6.7	7.3	8.2
		MCWBR	4.3	3.9	4.1	3.9	3.6	3.6	3.6	3.8	3.5	3.3	3.6	3.9
Clear-Sky Solar Irradiance	taub	0.421	0.448	0.464	0.481	0.473	0.444	0.455	0.456	0.428	0.412	0.434	0.422	
	taud	2.412	2.322	2.283	2.240	2.302	2.389	2.363	2.358	2.427	2.497	2.422	2.440	
	Ebn at Noon	890	875	863	837	827	841	835	847	882	898	871	880	
	Edh at Noon	118	133	140	144	132	120	123	126	119	110	116	113	
All-Sky Solar Radiation	RadAvg	5.13	5.52	5.54	5.38	4.54	4.20	4.15	4.27	4.36	4.27	3.95	4.48	
	RadStd	0.23	0.28	0.30	0.25	0.34	0.28	0.27	0.23	0.25	0.22	0.33	0.48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	-0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page